

NeuroScan AI v3.0 — Clinical Imaging Report

AI-assisted neuro-radiological screening (research use only)

Report ID	NS3-20260913-093731	Generated	13 Sep 2026 09:37
Source	brisc2025_test_00255_me_ax_t1.jpg	Input domain	full_image
Deployed arm	full_image	Backbone	EfficientNet-B0

PRIMARY AI IMPRESSION

AI impression	MENINGIOMA
Ensemble confidence	89.58%
Risk stratum	MEDIUM
Predictive uncertainty	sigma = 0.0268 (Low)
Branch consensus	100% of the three branches agree

CLASS POSTERIOR

Class	Ensemble	Standard	MC dropout	TTA	MC sigma	
glioma	4.40%	4.54%	5.08%	2.35%	0.0289	
meningioma	89.58%	89.71%	88.00%	94.30%	0.0446	SELECTED
no_tumor	4.68%	4.62%	5.52%	2.18%	0.0234	
pituitary	1.34%	1.13%	1.40%	1.17%	0.0105	

Fusion weights 0.00 / 0.75 / 0.25 (standard / MC dropout / TTA) were selected by grid search on the validation split; the held-out test split took no part in choosing them.

LESION GEOMETRY (Attention U-Net)

Area	2652 px ² (6.16% of the crop)	Circularity	0.745
Centroid	(164, 137)	Quadrant	Lower-Right

INTERPRETATION

Imaging features are consistent with meningioma, typically a slow-growing extra-axial lesion arising from arachnoid cap cells. A dural tail and homogeneous enhancement are characteristic. Neurosurgical consultation is indicated.

Recommendation: Neurosurgical consultation; contrast-enhanced MRI and visual field assessment.

MODEL PROVENANCE (measured, not asserted)

Dataset	BRISC2025 (briscdataset/brisc2025) — 4211 train / 744 val / 995 held-out test
Test split	Official BRISC2025 test folder, untouched during training
Segmentation	Attention U-Net on annotated masks — test Dice 0.8400, IoU 0.7756
Deployed classifier	full_image — test accuracy 98.59%, macro F1 98.62%
With ensemble	test accuracy 98.79%, macro F1 98.84%, kappa 0.9835
Controlled contrast	full-image arm macro F1 98.62% vs ROI arm 97.56%; McNemar p = 0.0433
Explainability	Grad-CAM on the last backbone convolution, computed on the same input domain the model was trained on
Uncertainty	MC dropout, 30 stochastic passes (BatchNorm frozen)

IMPORTANT: this is an experimental research system. This report is not a medical diagnosis, has not been reviewed by a licensed clinician, and must not be used to direct patient care. Model outputs are derived from a single 2D slice and must be correlated with the full study, clinical history and expert radiological interpretation. The authors accept no liability for decisions based on this output.